Beta regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$
\begin{aligned}
y_i \mid \mu_i,\, \sigma_i &\sim \mathrm{Beta}(\mu_i\sigma_i,\, (1-\mu_i)\sigma_i) \\
\mathrm{logit}(\mu_i) &= \beta_0 + \beta_1\, x_i \\
\log(\sigma_i) &= \gamma_0
\end{aligned}
$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$
\begin{aligned}
\mathbf{y} \mid \boldsymbol{\mu},\, \boldsymbol{\sigma} &\sim \mathrm{Beta}(\boldsymbol{\mu},\, \boldsymbol{\sigma}) \\
\mathrm{logit}(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta} \\
\log(\boldsymbol{\sigma}) &= \mathbf{Z}\boldsymbol{\gamma}
\end{aligned}
$$

3. Worked observation (i = 1)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$
\underbrace{\mathrm{y}_1}_{\text{observed}=0.175} \;=\; -0.824 + -0.0874 \cdot 1.43 \;+ 1.13 = 0.175
$$

Predicted $\hat{\mu}_1 = -0.95$; residual $\hat{\varepsilon}_1 = 1.13$.

And the σ submodel at $i = 1$:

$$
\log \hat{\sigma}_1 \;=\; -1.04 \;\Rightarrow\; \hat{\sigma}_1 \;=\; \exp(-1.04) \;\approx\; 0.353
$$

Stacking the same response equation for all $n = 100$ observations:

$$
\underbrace{\begin{bmatrix} 0.175 \\ 0.324 \\ 0.146 \\ 0.357 \\ 0.148 \\ \vdots \\ 0.174 \\ 0.201 \end{bmatrix}}_{\mathbf{y}_{100\times1}\ \text{(observed)}}
=
\underbrace{\begin{bmatrix} 1 & 1.43 \\ 1 & -0.651 \\ 1 & -0.207 \\ 1 & -0.393 \\ 1 & -0.32 \\ \vdots & \vdots \\ 1 & -0.164 \\ 1 & 0.421 \end{bmatrix}}_{\mathbf{X}_{100\times2}}
\underbrace{\begin{bmatrix} -0.824 \\ -0.0874 \end{bmatrix}}_{\hat{\boldsymbol{\beta}}_{2\times1}\ \text{(estimated)}}
+
\underbrace{\begin{bmatrix} 1.13 \\ 1.09 \\ 0.952 \\ 1.15 \\ 0.944 \\ \vdots \\ 0.984 \\ 1.06 \end{bmatrix}}_{\hat{\boldsymbol{\varepsilon}}_{100\times1}\ \text{(residual)}}
$$

And the σ submodel (no observed counterpart – σ's job is to describe the spread of $\hat{\boldsymbol{\varepsilon}}$):

$$
\log \underbrace{\begin{bmatrix} 0.353 \\ 0.353 \\ 0.353 \\ 0.353 \\ 0.353 \\ \vdots \\ 0.353 \\ 0.353 \end{bmatrix}}_{\boldsymbol{\sigma}_{\,100\times 1}} = \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}}_{\mathbf{X}_{\sigma,\,100\times 1}} \underbrace{\begin{bmatrix} -1.04 \end{bmatrix}}_{\boldsymbol{\gamma}_{\,1\times 1}}
$$